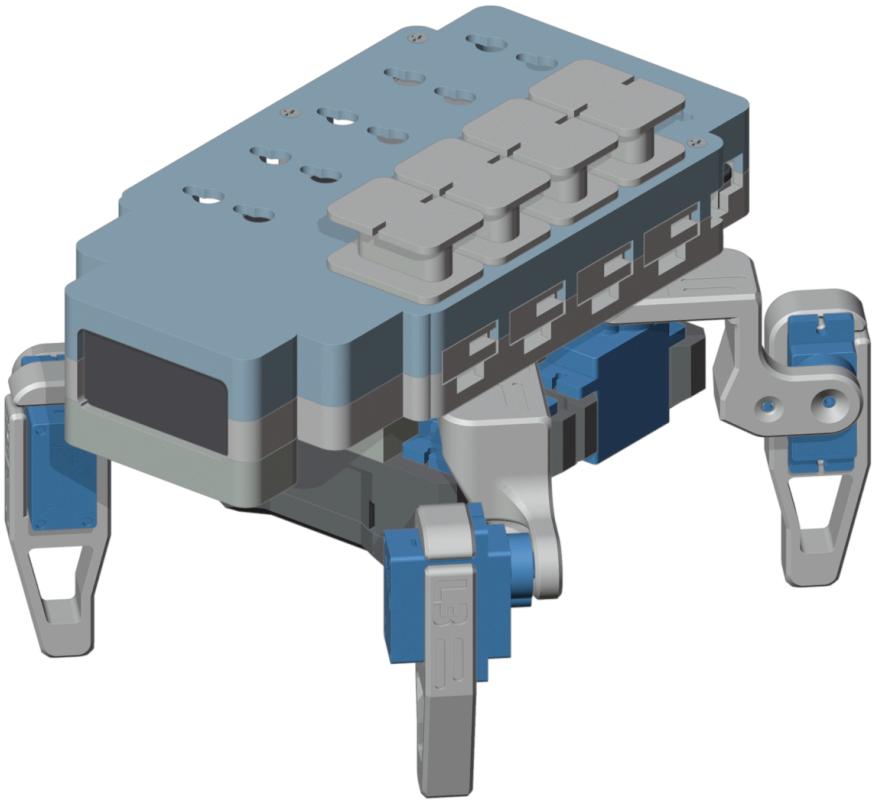
Atech-crab

Assembly guide · print files v6 (holes for the wires) · 9 November 2026



Atech-crab is Sesame, Dorian Todd's open-source quadruped (Apache-2.0), with an Atech board as its brain. Its frame, servos and legs stay as Sesame makes them, and a new deck and lid from Atech Atelier ("Sesame walker" 0.1.3) carry an Atech 14-port board flat on top, with the screen in a visor at the front. The eight MG90S servos plug into the board's ports 2–5 on jumper wires.

New in v6: a way through for every wire

- **8 lead slots through the internal frame**, one per servo, from the wire channels underneath up into the spine.
- **8 lead holes in the spine's front and back walls**, one per servo: the other way up, if you prefer it.
- **Slots at every side port**, 2–6 and 10–14, half in the lid and half in the deck, each with a lead notch under it (v4, v5).
- **20 keyholes in the lid**, two by each side port, for the wire bobbins.
- **16 small wire parts**: 4 port jumper holders, 8 servo plug sleeves and 4 wire bobbins.

> **Not built yet.** These pictures come from the CAD, which passes Atelier's fit check, and the wire parts were checked against it. Nobody has printed and assembled v6. Test-fit the deck on the frame, with the servos in, and try one jumper holder in a port, before you print a second set.

What you need

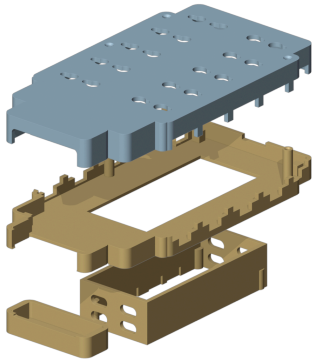
Printed: 13 parts + 16 wire parts

Bambu Lab printers: one file prints everything, e.g. `atech_crab_everything_pla_h2dp.gcode.3mf` for the H2D Pro (atech/print/gcode/). Other printers: atech/print/README.md.

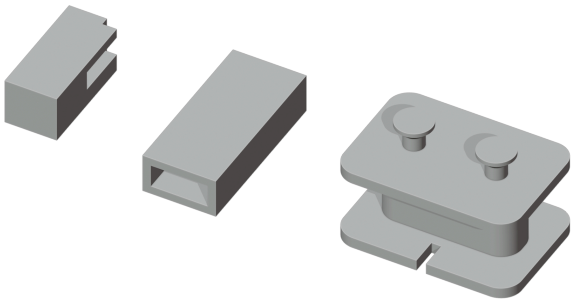
Part	Qty	From
Internal frame, with lead slots	1	Sesame, v6
Bottom cover	1	Sesame
Hip legs (femurs) L1, L2, R1, R2	4	Sesame
Feet L3, L4, R3, R4	4	Sesame
Deck part 1 (plate)	1	Atelier, v5
Deck part 2 (spine and head), with lead holes	1	Atelier, v6
Lid, with keyholes	1	Atelier, v6
Port jumper holder	4	wire part
Servo plug sleeve	8	wire part
Wire bobbin	4	wire part

Hardware

Item	Qty
MG90S servos (metal gear, 180°)	8
Atech 14-port board	1
Atech USB-C module (port 8)	1
Atech motion sensor module, icm40608 (port 13)	1
Atech screen module and its cable (ports 10+11)	1
M2×8 pan-head screws (deck 4, bottom cover 2)	6
M2×10 countersunk screws (lid)	3
Male-to-male jumper wires (3 a port)	12+
5 V 3 A supply for the servos	1
Superglue (CA)	—



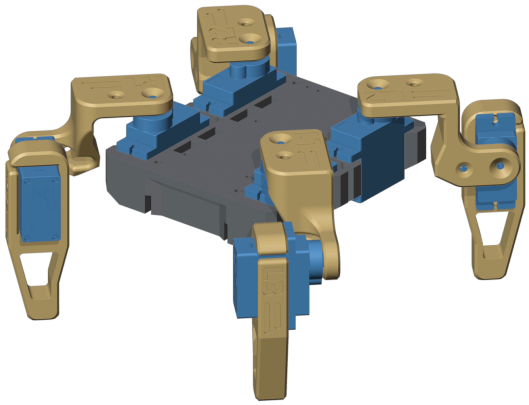
The three Atelier parts: lid (top), deck part 1, deck part 2.



The wire parts: a port jumper holder, a servo plug sleeve and a wire bobbin (pegs up, as it prints).

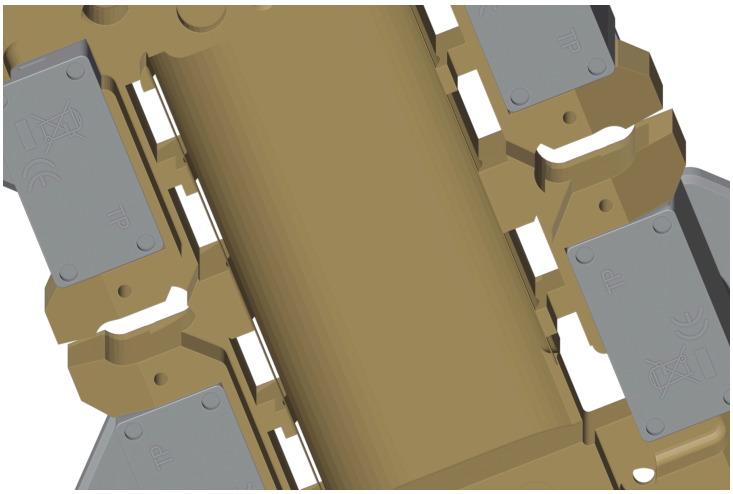
In the step pictures, the parts you add in that step are in colour; the parts already in place are pale grey; printed parts you add are amber.

1 Build the base



1. Put the four hip servos into the frame, and the four knee servos into the femurs, as in Sesame's own guide (github.com/dorianborian/sesame-robot).
2. Fit the femurs and the feet onto the servo horns.
3. Leave the bottom cover off for now: the deck screws go in from inside the frame.

The leads: up through the frame

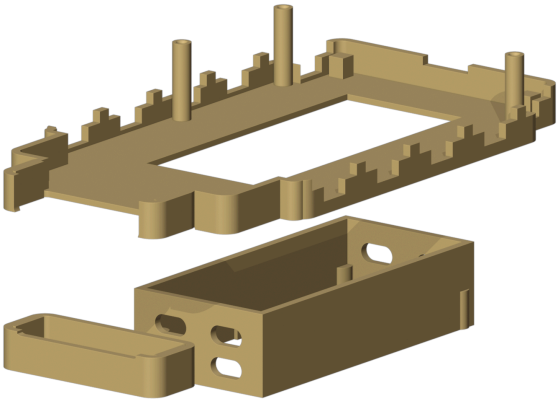


The frame from below (amber). Each wire channel has two 9 × 3.4 mm lead slots going up into the spine, one per servo.

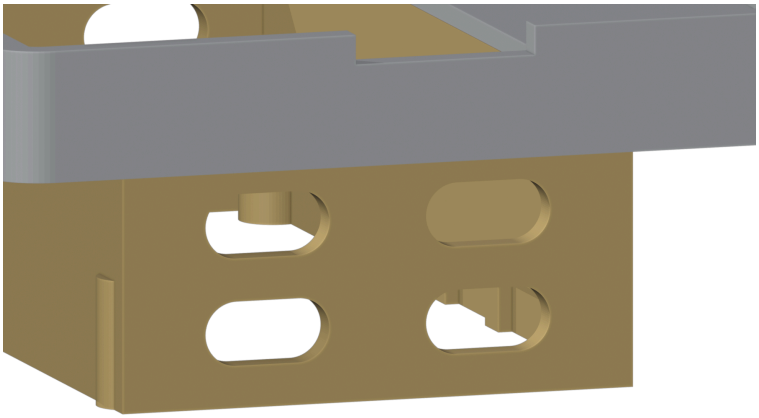
1. With the bottom cover off, push each servo's plug up through a lead slot in the frame, then press the lead into the channel underneath, as Sesame's guide does.
2. Inside, the plug comes up beside the spine. The spine wall over each slot has a 1.1 mm notch on its inside, so the plug gets past it.

There is a second way in: the lead holes in the spine's end walls (step 2). Use either route for each lead.

2 Glue the deck



1. Dry-fit deck part 2 (spine and head) under deck part 1. The 0.6 mm lip fits the recess, so it only goes one way.
2. Glue it with superglue (CA) and hold it until it sets.

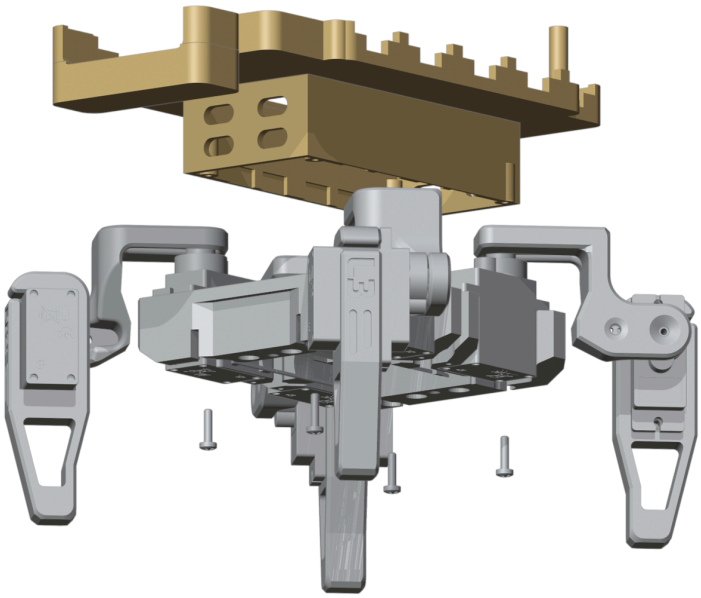


The spine's back wall: four lead holes, 9 × 4.5 mm with round ends, so a servo plug passes through. The front wall has the same four.

The other route: the spine's lead holes

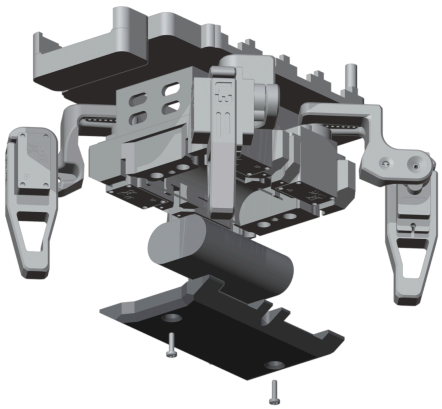
- **Front wall:** the front legs' leads. **Back wall:** the rear legs' leads.
- On each leg's own side of the wall, the **hip** lead goes in the **lower** hole and the **knee** lead in the **upper** one.
- The side walls have no holes: the femurs pass 0.2 mm outside them.

3 Deck onto the frame



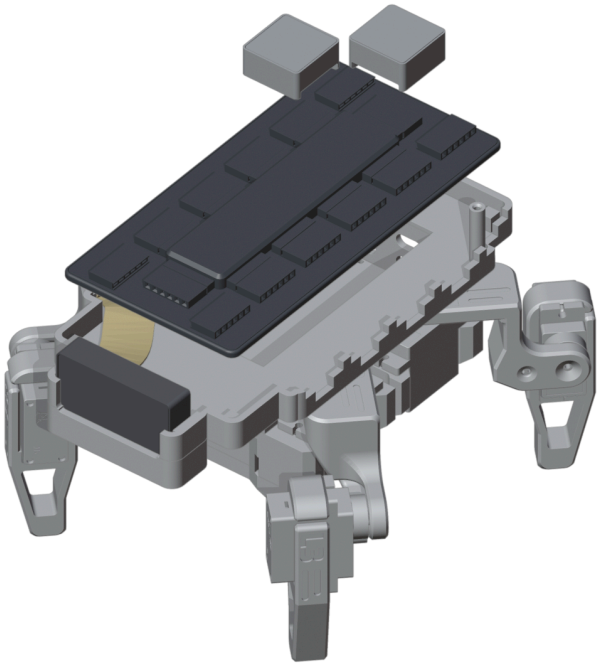
1. Lower the deck onto the frame with every servo lead inside the spine: up from the frame's lead slots, or threaded through the spine's end-wall holes.
2. Pull each lead up the spine to the 5 mm gap under the board.
3. From inside the frame, through its open bottom, drive 4 × M2×8 pan-head screws up into the deck's bosses.

4 Battery and bottom cover



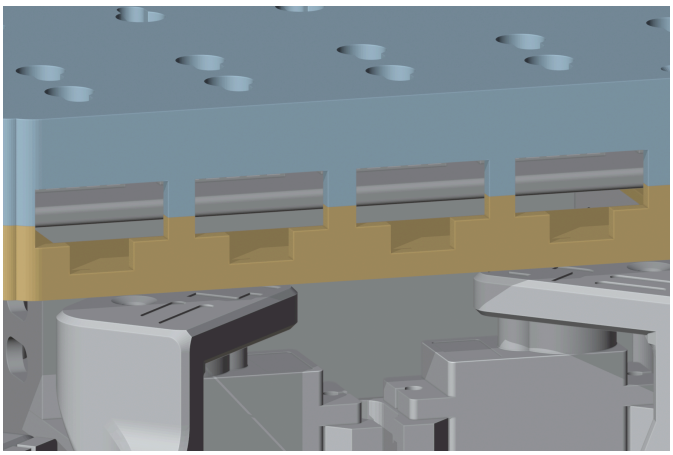
1. Check that every lead lies in its channel and none is pinched at the edge of the frame.
2. Put Sesame's battery into the frame.
3. Close the bottom with the bottom cover: 2 × M2×8 pan-head screws.

5 Board, modules and screen

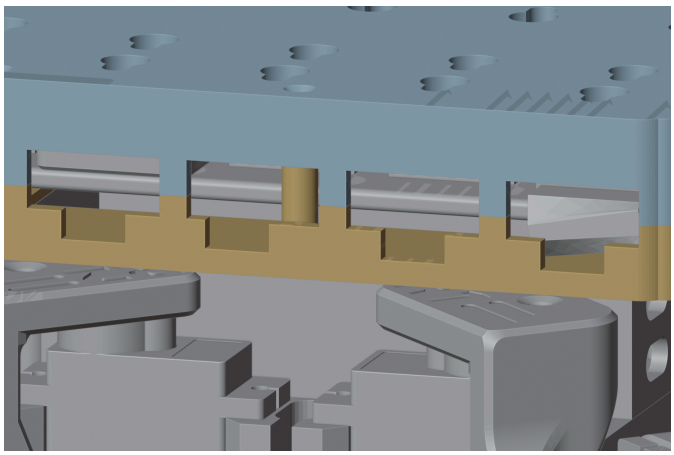


1. Lay the servo leads in the 5 mm gap under the board, each toward its port (table in step 6). A lead leaves through the notch under its port's slot: ports 2–6 on the right, 10–14 on the left, whichever is shorter.
2. Set the board on the four posts in the tray, ports 1–6 on the right, port 8 at the back.
3. Slide the USB-C module into port 8 (its socket faces the back wall), and the motion sensor into port 13.
4. Stand the screen upright in the head, behind the window, connector edge down. Its two ribbons loop down into the right-hand bay, run flat under the board and plug into ports 10 and 11.

6 Wire the servos

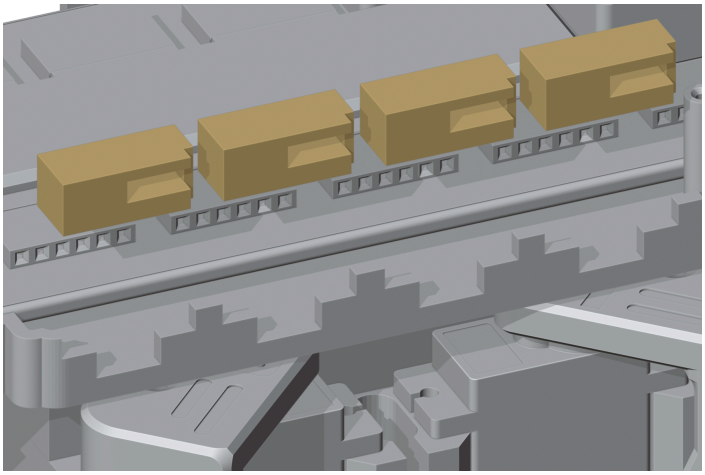


The slots at ports 2–6, half lid, half deck, and the lead notches under them.



The same on the left, at ports 10–14. Port 12's notch sits 2 mm back, clear of a lid-screw post.

Port	Leg	Line A (pin 5)	Line B (pin 6)
2	front left	L1 hip	L3 knee
3	front right	R1 hip	R3 knee
4	back left	L2 hip	L4 knee
5	back right	R2 hip	R4 knee

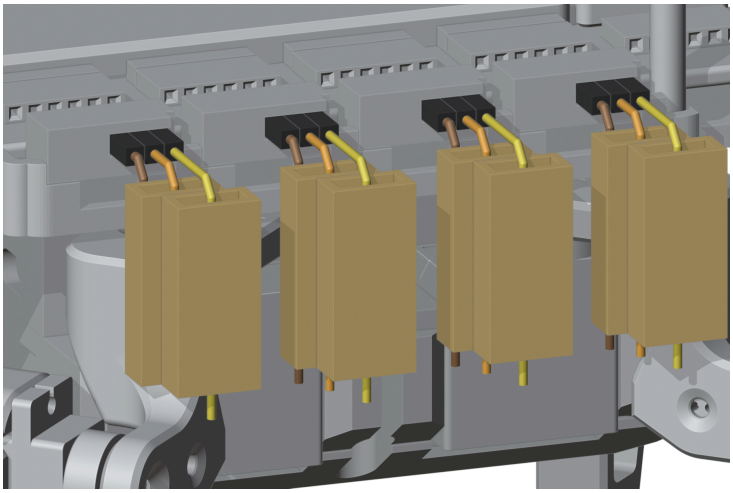


A jumper holder drops onto each of ports 2–5, its tongue in the port's slot.

Jumper holders

1. Per port, push three jumpers into the holder's channel: ● GND (pin 4), ● line A (pin 5), ● line B (pin 6).
2. Plug the holder onto the port as one piece. Its solid half covers pins 1–3 (+12 V, 5 V, 3.3 V), toward the front of the robot, so no jumper can reach them. Upside down it does not go in.
3. The same part, turned round, fits ports 10–13.

6 (continued)



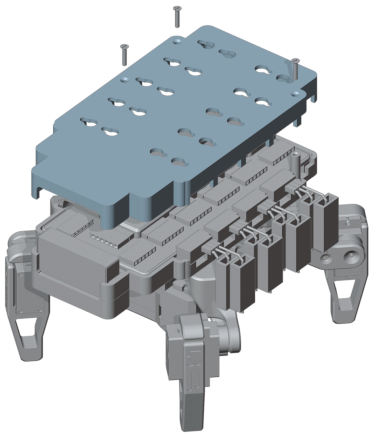
Jumpers out of the holders, a sleeve on each servo joint (an illustration: the servo leads and the power wires are not drawn).

Join the servos

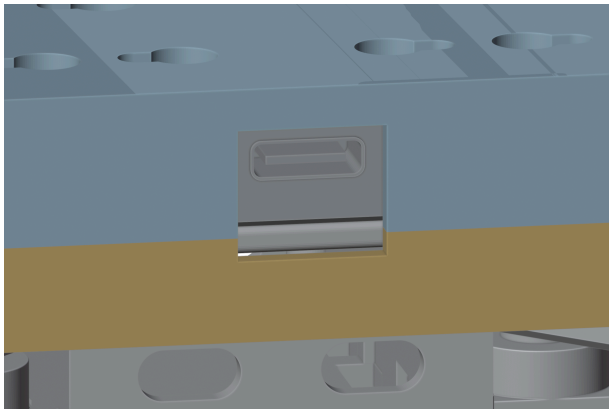
1. Thread a plug sleeve onto each signal jumper, then plug the jumpers into the servo's 3-pin plug: the servo's orange (or yellow) wire to its signal jumper, line A for the hip and line B for the knee.
2. Slide the sleeve over the joint, so the plug cannot pull apart.
3. Servo power: every servo's red wire to the 5 V supply's +, every brown wire to its –, and the supply's – to a GND jumper.

+12 V sits next to +5 V. Port pins: 1 +12 V, 2 +5 V, 3 +3.3 V, 4 GND, 5 line A, 6 line B. The holder covers pins 1–3 when it goes in the right way round, with its solid half to the front. If you wire without a holder, measure first: with the board on USB, the pin at 3.3 V is pin 3, and its neighbour that beeps to the USB-C shell is GND (pin 4). Never connect pin 1.

7 Close the lid

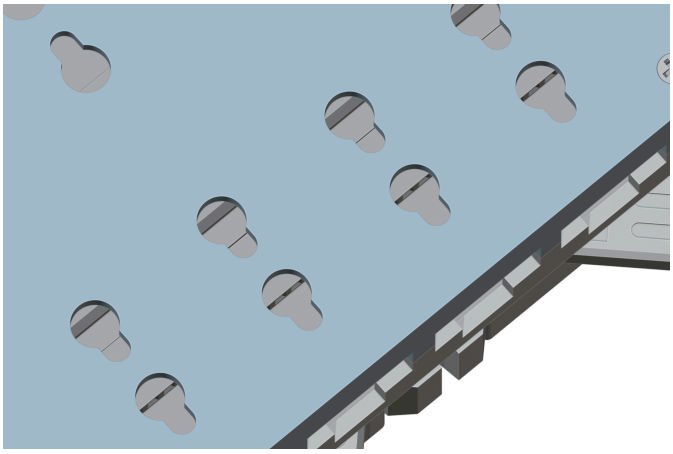


1. Check that no ribbon or lead is pinched between the deck and the lid, and that each holder's tongue sits in its slot: the lid holds it there.
2. Fit the lid and drive 3 × M2×10 countersunk screws into the bosses in the side bays.

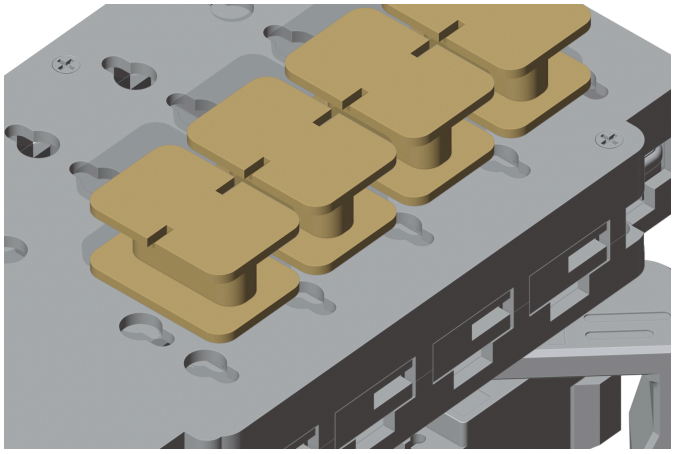


The USB-C opening at the back, 13 × 11.5 mm, across the seam.

Wire bobbins: the slack on the lid



Two keyholes by each side port.



A bobbin by each of ports 2–5, one per leg.

1. Wind a port's slack round its bobbin's core: both servo leads and the port's jumpers. Pinch each end in a slit in the top flange.
2. Drop the bobbin's two pegs into the port's keyholes and slide it 4 mm outward, toward the port: the wires' pull keeps it locked. To take it off, slide it inward and lift.
3. At port 12 (and nearly at port 6) the bobbin covers a lid screw: take the bobbin off before you open the lid.

8 First run

1. Plug a USB-C data cable into the back, and switch on the servos' 5 V supply.
2. Flash the walker: in `atech/firmware`, run `atech upload .`
3. Check the board: `atech check`. Then try one move with `atech send spider_go wave`, holding the robot off the table.

The walker firmware and its gaits have been run only in simulation (an emulated board in MuJoCo), and the MG90S servo driver not yet with MG90S servos on a board. A servo that only twitches is short of power: use the separate 5 V supply.

